Michał Kwak

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EDUCATION

BSc Computer and Data Science, University of Helsinki **09/2024 - 06/2029 (expected)**

- GPA 4.86 / 5
- Thesis: *Evaluation Practices in Machine Learning for Hypoglycemia Prediction from Continuous Glucose Monitoring Data* (supervisors: Assoc. Prof. Luigi Acerbi, Dr. Herilalaina Rakotoarison)

RESEARCH

Hypoglycemia prediction from CGM data, University of Helsinki **09/2026 - present**

- Compared published Random Forest, XGBoost and BiLSTM approaches for predicting glucose below 70 mg/dL from continuous glucose monitoring data for type 1 diabetes patients
- Ran a controlled comparison of the models on the public CITY dataset with identical data, labels and splits
- Extending the work with participant-level cross-validation, external validation on OhioT1DM and an alarm-level metric (false alarms per patient-day)

Gaussian state tomography of Hawking/Unruh toy radiation, team research project **05/2026 - present**

- Built a Python simulation and tomography pipeline modelling the radiation as a two-mode squeezed state
- Designed a modular simulation engine supporting joint, homodyne and heterodyne measurements

EXPERIENCE

Co-founder and Developer, Syllabuss (Helsinki Incubators) **09/2025 - present**

- Co-founded a study-planning platform that helps students navigate degree structures
- Built backend tools, including a scraper for university course data, using Git/GitHub in a team
- Ran student outreach across faculties and used the feedback to guide the product

Teaching Assistant, Computer and Internet, University of Helsinki **01/2026 - 05/2026**

- Helped students debug programming problems in exercise sessions and explained technical concepts to international groups

International Tutor Coordinator, University of Helsinki **01/2026 – 09/2026**

- Led the recruitment of international tutors and worked with the faculty to plan the 2026 Orientation Week

Ambassador, Study in Finland **03/2026 - present**

- Guided prospective students and represented Finnish higher education at international events

PROJECTS

Steam Decline Watch GitHub

- Collaborated in a team on a project predicting Steam games' collapses
- Built and benchmarked a calibrated random forest with grouped CV: ROC-AUC 0.84 vs 0.59 baseline with the monthly backtest showing the top 10% watch list caught 56% of collapses
- Deployed the model as a live website with an interactive dashboard

Finnish Municipal Economic Resilience Index GitHub

- Built a Python pipeline on the Statistics Finland API scoring 308 municipalities over 1995-2024
- Measured recession resilience and shipped it as a CLI with uv, pytest, Ruff and GitHub Actions CI

Face recognition with eigenfaces, AI and Algorithms lab course, University of Helsinki GitHub

- Implemented eigenfaces from scratch in Python/NumPy using a Jacobi solver with 87.5% accuracy
- Tested with pytest (96% coverage) and documented the work independently in weekly reports

SKILLS

Programming: Python (pandas, NumPy, scikit-learn, matplotlib), SQL, R, C#, JavaScript (React), HTML/CSS
Tools: Git/GitHub, GitHub Actions CI, Linux, Jupyter, Unity, Claude Code
Languages: English (C2), Polish (native), Finnish (A1), Japanese (JLPT N5)

AWARDS

Małopolska Regional Scholarship for top STEM students 2019 and 2021

Finalist, 10th National Logic Competition | John Paul II Catholic University of Lublin 2022